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Top Skills

Claude Code
Vibe Coding
OpenAI Codex

Certifications

Mechanical Design part(2)
SOLIDWORKS: Animations
Certified SOLIDWORKS Mechanical Design Associate
Introduction to Self-Driving Cars
Ansys Static Analysis - FEA Basics

Patents

Detecting a Lane Change Maneuver of Ego Vehicle and Differentiating Ego Driving on Lane Boundary and Ego Lane Change Maneuvers to Improve ADAS Feature Performance and V2X
ACC Target Selection for Crossing Objects
Filtered Boosting of Yaw Rate signal for Enhanced Ego-Vehicle Path Prediction for Active Safety End-User Functionalities

Ahmed Ezz El-Deen

ADAS Feature Owner - System Engineer at Valeo
Cairo, Egypt

Experience

Valeo

4 years 2 months

System Engineer

March 2025 - Present (1 year 7 months)

As the Feature Owner and System Engineer for Adaptive Cruise Control (ACC), I am responsible for the end-to-end ownership of the ACC platform functionality within the vehicle's driver assistance system that affects globally for ACC system. Building on my prior experience as an ADAS Software Engineer, I now lead the definition, alignment, and delivery of system-level behavior that ensures robust, safe, and intuitive longitudinal control for the driver

Key Responsibilities:

- Own the system-level definition and evolution of the ACC feature across global vehicle platforms (Europe, China)
- Translate customer and regulatory requirements into clear functional specifications and use cases and system requirements
- Align cross-domain stakeholder inputs (software, hardware, sensor integration, validation, HMI) into cohesive system requirements and design
- Support integration, test strategy, and validation of ACC across ViL, SiL, HiL
- Own first-line problem diagnosis for complex system faults: fault sensor calibration/setup, communication issues, decision making issues and control anomalies; coordinated fixes across software and hardware and testing team.
- Enhancing ACC behavior in complex scenarios and supporting cross-functional teams in issue resolution and continuous improvement of ACC target object selection to follow KPIs
- Delivery of production ready system which enhanced my familiarity to ISO 26262 and SOTIF practices
- Responsible for ACC test campaign for Euro NCAP ADAS tests, ensuring system behavior aligned with regulatory and consumer and costumer safety

benchmarks, providing detailed feedback and root-cause analysis from Euro NCAP-style scenario testing, driving improvements in perception, decision-making, and control modules

ADAS Software Engineer

September 2023 - March 2025 (1 year 7 months)

Cairo, Egypt

Focused on a role in the development and refinement of the Adaptive Cruise Control (ACC) system. I was heavily involved in the logic design and decision-making algorithms for target object selection in dynamic driving environments:

- Led the development of complex decision-making algorithms for ACC, specifically targeting the logic behind selecting which object (vehicle, cyclist / motorbikes, trucks, etc.) in the environment the system should follow.
- Successfully passed multiple customers' Start of Production (SOP) milestones, delivering a fully functional ACC system that met stringent safety and performance requirements. This involved extensive testing, validation, and fine-tuning of the system to meet both regulatory standards and customer expectations.
- Demonstrated leadership in troubleshooting and solving complex system issues, driving continuous improvements and enhancements to the ACC functionality

Junior ADAS Software Engineer

August 2022 - September 2023 (1 year 2 months)

Cairo, Egypt

As a junior ADAS algorithm Engineer, I played a key role in the development and testing of high-speed Advanced Driver Assistance Systems (ADAS). My work primarily focused on ensuring robustness and performance of the system.

- Conducted comprehensive SIL testing for high-speed ADAS systems, ensuring functionality and reliability under a wide range of simulated driving scenarios following system requirements
- Contributed to the development of lane centering system (LCS) functionality, which is critical for maintaining vehicle stability and safety in high-speed environments

- Refined algorithms for ego motion states and ego path prediction, crucial components for delivering advanced driver support functionalities such as Adaptive Cruise Control (ACC) and lane centering. This involved ensuring that the vehicle's motion and future trajectory were predicted in real-time

- Worked extensively on lane change assist features, including Blind Spot Detection (BSD) and Close Vehicle Warning (CVW) and collision avoidance features such as Rear Collision Warning (RCW), and briefly on Rear and Front Cross Traffic Warning (RCTW/FCTW). These functions were integral in improving situational awareness and minimizing the risk of collisions, contributing to overall driver safety.

COM-IoT Technologies

Research And Development Intern

March 2022 - August 2022 (6 months)

Dubai, United Arab Emirates

Worked on multiple tasks using python and C, worked with Robotics framework (ROS) and Docker.

- contributed to the POC of I-Park solution using LIDAR:

- Developed a real-time live, continuous update of parking lot vacant spots, occupied spots. -Development of a smart updating GUI to match the real-time status of the parking lots either vacant or occupied.

ASU Racing Team

Perception Engineer at Formula AI

September 2021 - July 2022 (11 months)

Ain Shams University

My Graduation Project is used to detect CONES accurate position relative to the vehicle using LiDAR technology

- Using python bindings for PCL
- RANSAC algorithm for ground plan removal
- Kd-tree for Clustering
- Fitting each cluster on CONE equation to filter cones from each cluster given
- Using SORT 3D to track Cones predictions and update the tracked cones' accurate position
- Using ROS to integrate between packages and custom messages for cones position

CONCORD FOR ENGINEERING & CONTRACTING

Mechanical Engineering Intern

September 2019 - October 2019 (2 months)

El Qahera El Gididaa, Cairo, Egypt

Mechanical Site (Water Lift Station) at new cairo got to learn about:

- Centrifugal pumps and how they are Controlled
- Connections using pressure Valves, Safety Valves
- Controlling the Flow of the water through the pipes

ENPPI

Internship Trainee

July 2019 - July 2019 (1 month)

Cairo Governorate, Egypt

Program specialty was mechanical engineering - 60hrs-

What i have managed to learn:

- .Material selection for process units including corrosion allowance calculations
- .Intro to material selection
- .ENPPI scope as EPC contractor for turnkey projects
- . Intro to pumps types, applications and design
- .Intro to air compressor packages, types and applications
- .HVAC principles
- .Intro to pressure vessels types, components &mechanical design
- .Intro to storage tanks classification and components
- .Intro to steam boilers

Education

Ain Shams University

Graduate, Mechatronics, Robotics, and Automation

Engineering · (2017 - 2022)